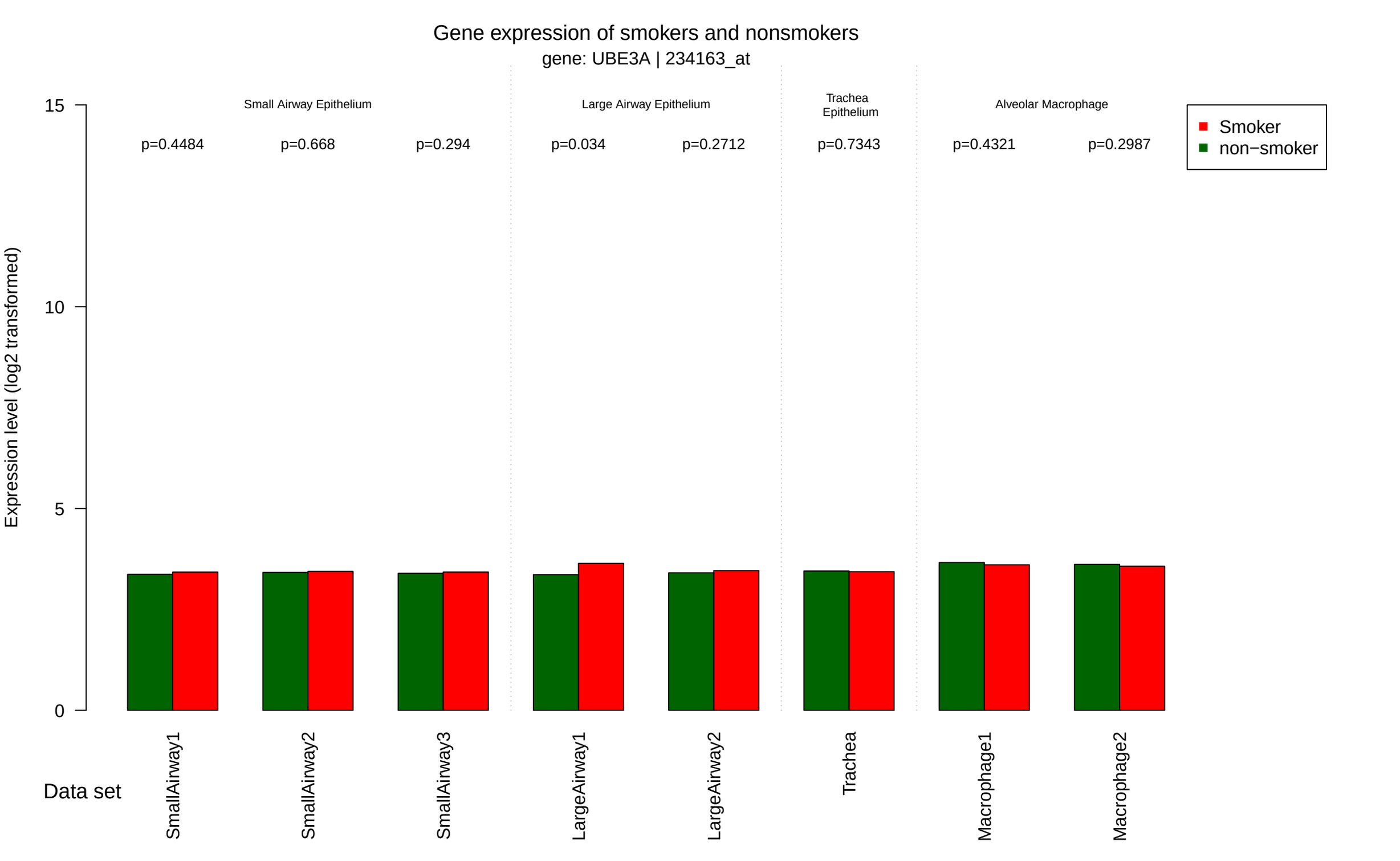
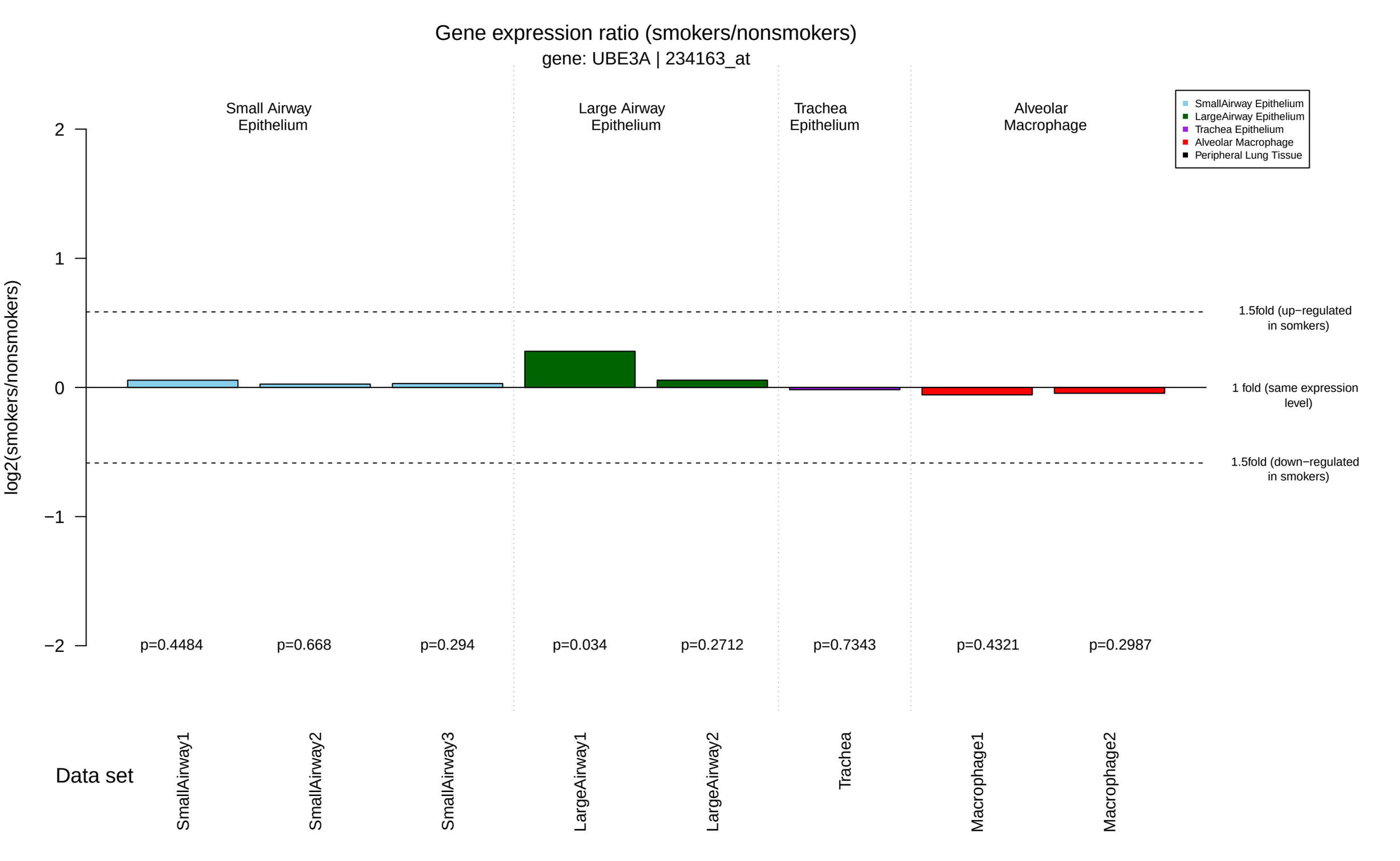
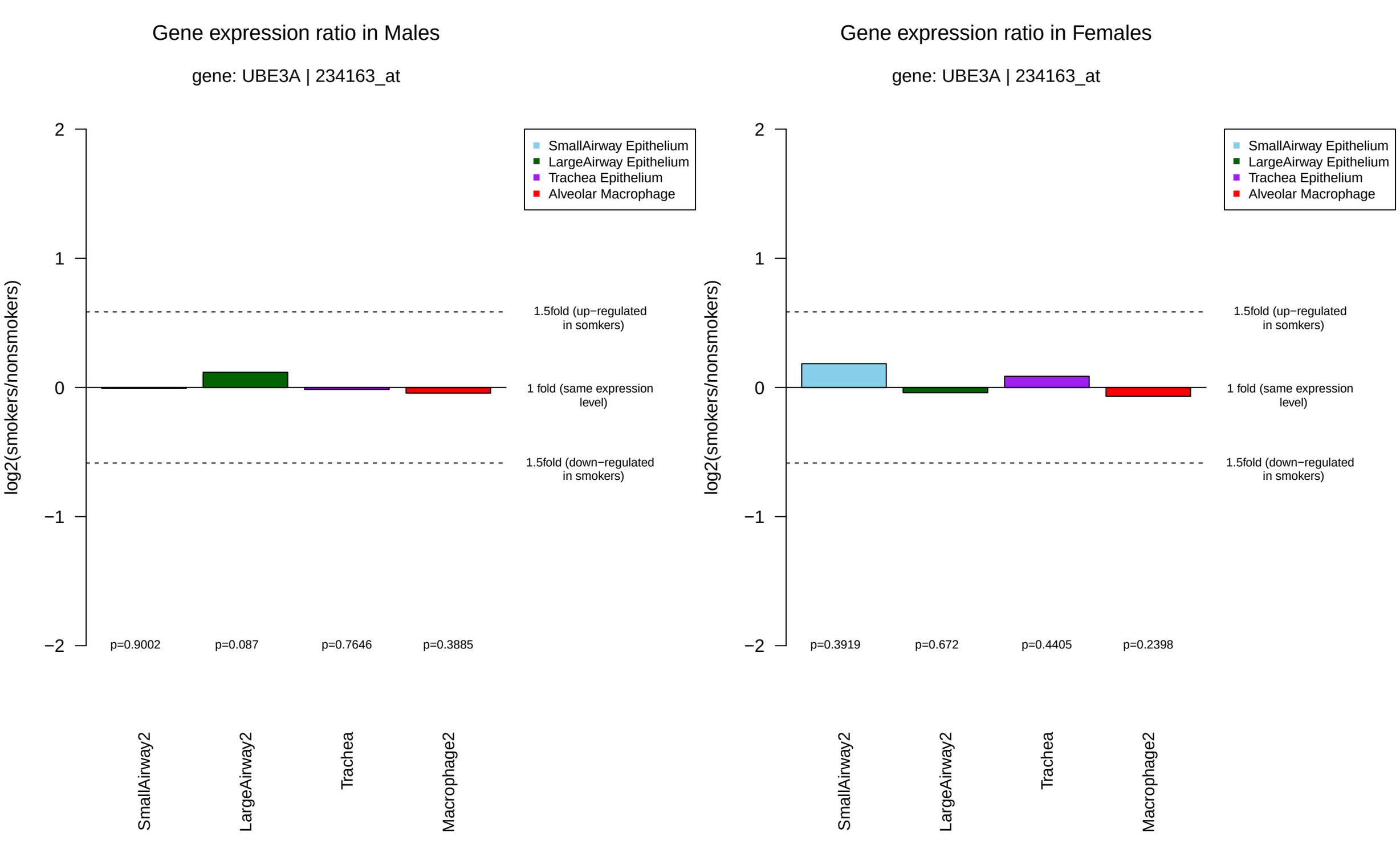
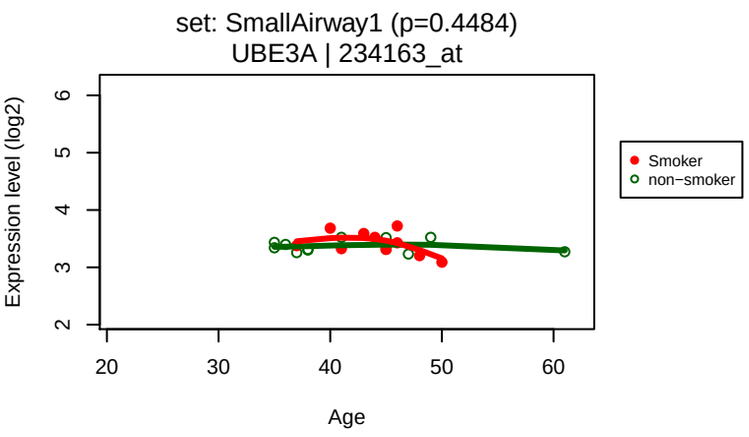
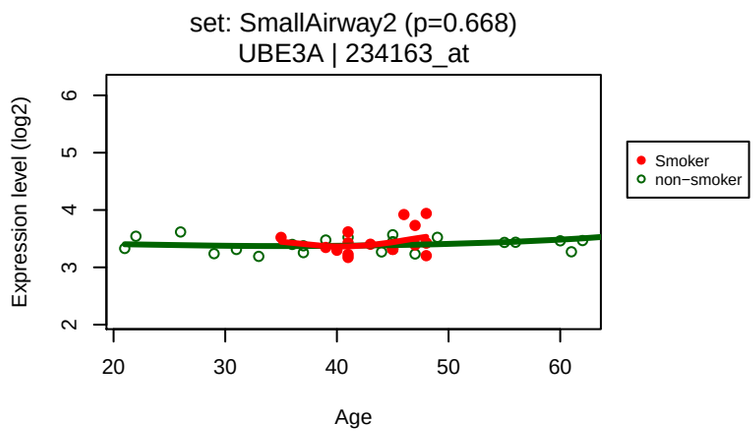
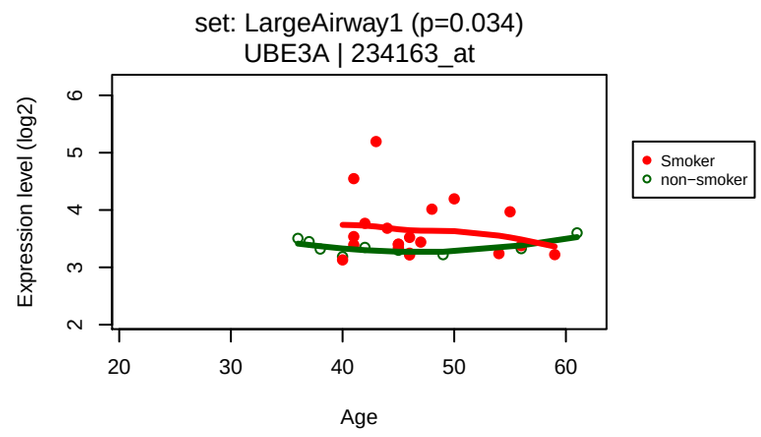

Gene expression of smokers and nonsmokers
gene: UBE3A | 234163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4484
p=0.668
p=0.294
p=0.034
p=0.2712
p=0.7343
p=0.4321
p=0.2987
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

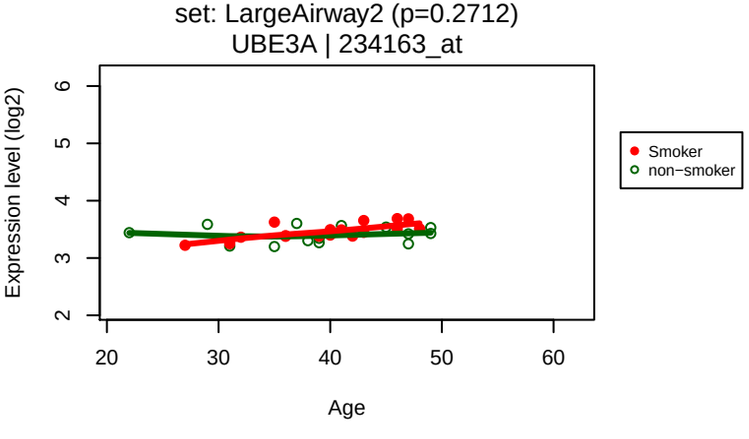

Gene expression ratio (smokers/nonsmokers)
gene: UBE3A | 234163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4484
p=0.668
p=0.294
p=0.034
p=0.2712
p=0.7343
p=0.4321
p=0.2987
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

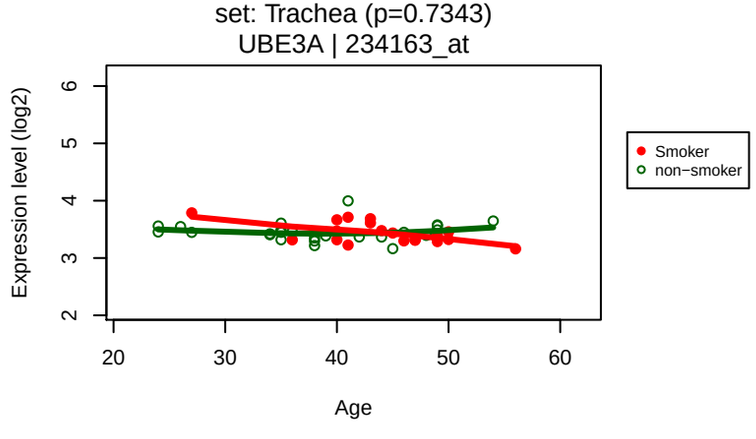

Gene expression ratio in Males
gene: UBE3A | 234163_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9002
p=0.087
p=0.7646
p=0.3885
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: UBE3A | 234163_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3919
p=0.672
p=0.4405
p=0.2398
SmallAirway2
LargeAirway2
Trachea
Macrophage2

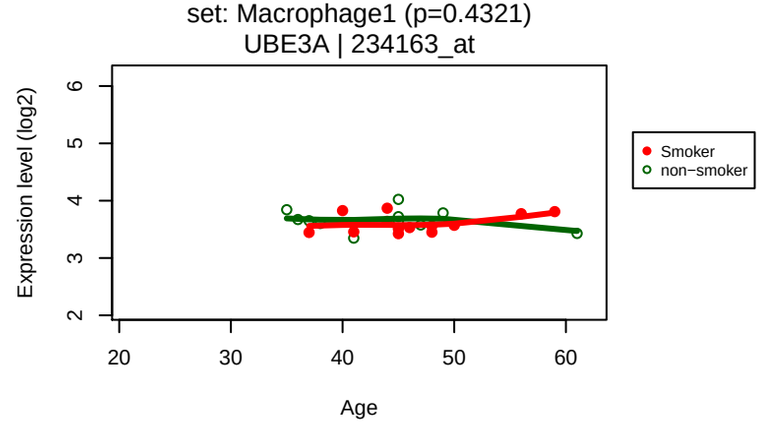

set: SmallAirway1 (p=0.4484)
UBE3A | 234163_at
Expression level (log2)
Age
Smoker
non−smoker

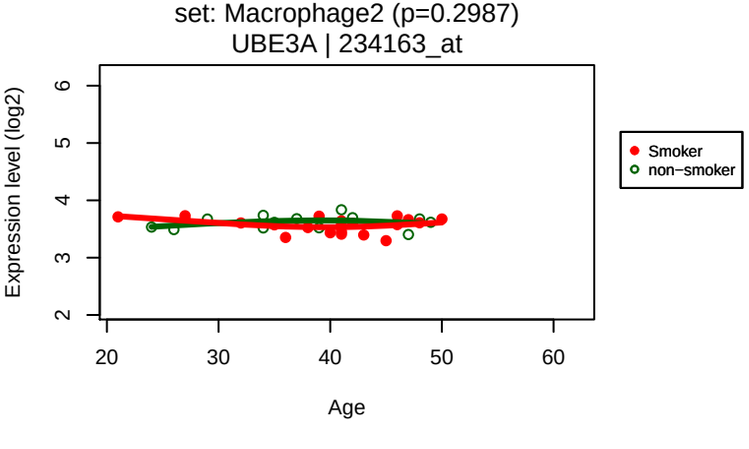

set: SmallAirway2 (p=0.668)
UBE3A | 234163_at
Expression level (log2)
Age
Smoker
non−smoker

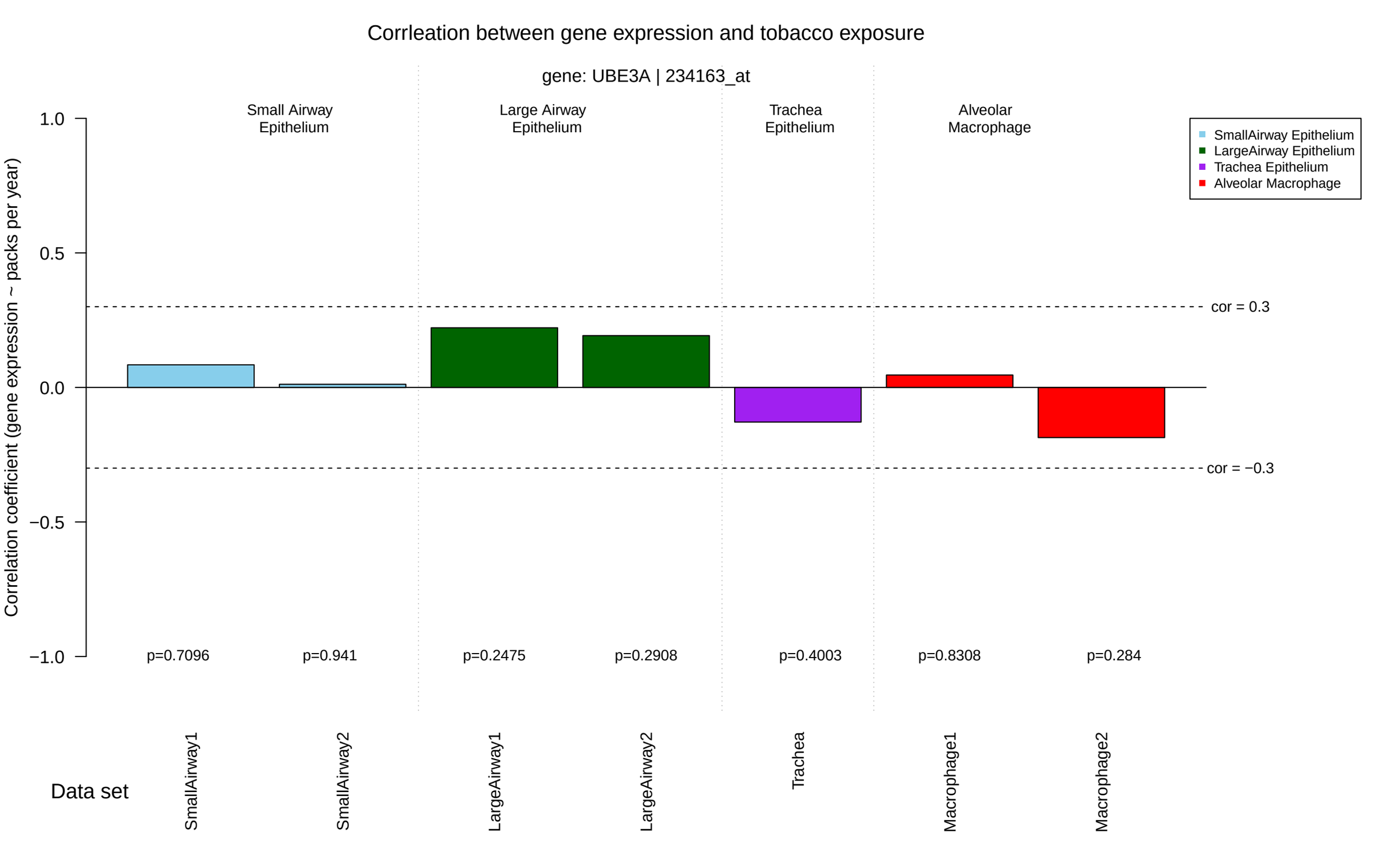

set: LargeAirway1 (p=0.034)
UBE3A | 234163_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2712)
UBE3A | 234163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7343)
UBE3A | 234163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4321)
UBE3A | 234163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2987)
UBE3A | 234163_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: UBE3A | 234163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7096
p=0.941
p=0.2475
p=0.2908
p=0.4003
p=0.8308
p=0.284
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2